Oluwatosin Ayeni

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Anthropic, London, UK

Dear Hiring Team,

I am applying for the Staff Cluster Infrastructure Engineer role at Anthropic London. Building and operating the GPU compute infrastructure that trains Claude — at the reliability and scale that frontier AI demands — is the most consequential infrastructure problem I know of, and it is exactly where I want to apply my experience.

Three specific achievements that map directly to this role:

- **Production GPU cluster infrastructure — end-to-end ownership:** I designed, provisioned, and operated a full production GPU cluster on Google Kubernetes Engine (NVIDIA L4 24GB) using Terraform IaC: node pools, networking, IAM, GCS storage, cluster security policies. Kyverno pod security admission, NetworkPolicy default-deny across namespaces, cert-manager TLS, KEDA autoscaling on GPU queue depth (not CPU — GPU pods sat at 5–15% CPU when saturated). Argo CD GitOps with Argo Rollouts canary deployments. LIVE over 5.3 days: 723 requests, 1.25M+ tokens, $152.82 infrastructure cost — this is full cluster lifecycle ownership, not a demo.
- **GPU observability and reliability at the hardware level:** I built a 27-target Prometheus stack with DCGM Exporter for GPU telemetry, AlertManager with 6 alert rules (memory pressure, pod restart loops, SLO breach), Grafana dashboards for cluster health, and Loki + FluentBit DaemonSet for structured log aggregation. At the CUDA level, I used NVIDIA Nsight Systems/ Compute to profile kernel execution on H100 NVL hardware: identified a memory bandwidth bottleneck in KV cache access patterns, achieving 3.8× throughput improvement and a 1,443× TTFT speedup via RadixAttention prefix caching at 131K-token context.
- **Infrastructure reliability under production load:** Via continuous batching pipeline redesign and kernel-level optimisation on RTX 4090, I achieved 28× LLM cost reduction and 18× latency improvement. I built *vllm-k8s-stack* — KV-cache-aware routing, HPA/KEDA autoscaling, hostPath weight caching (5 min cold load → 30 s cached), Prometheus/Grafana/AlertManager, and Argo CD GitOps — a cluster serving stack built for reliability, not just throughput.

The infrastructure challenges at Anthropic's scale — managing thousands of GPUs, maintaining cluster reliability during long training runs, ensuring network fabric health for all-reduce collectives, and keeping compute utilisation high — are the exact class of problem I have been building toward from first principles. I use Claude in production daily and I take the safety-reliability connection seriously: reliable infrastructure is what makes responsible AI possible.

I am planning to relocate to London and available to interview at your convenience.

Yours sincerely,

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